

**Faculty of Engineering and Computer Science
Department of Electrical and Computer Engineering**

**COEN / ELEC 490
CAPSTONE PROJECT**

MANUAL

**CONCORDIA UNIVERSITY
2026**

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Course Description

The ELEC/COEN 490 project is intended as a culmination project in our engineering programs. It is designed for students to gain significant design experience while applying integrated knowledge from several courses. It is also a means to practise project management, technical writing, and technical presentation and other soft skills.

Students work in groups under faculty supervision to solve complex interdisciplinary design problems typically involving communications, signal processing, control systems, electromagnetics, power electronics, software design, and/or hardware design. The project fosters teamwork between group members and allows students to develop their ability to carry out the work in various aspects of real engineering projects.

Tutorial: one hour per week, two terms. Equivalent laboratory time: nine hours per week, two terms.

NOTE: All written documentation must follow the Concordia Form and Style guide. Students are responsible for obtaining this document before beginning the project.

The two-term project is divided into four phases:

Phase 1: Project selection and planning: formation of the team, selection and definition of the project, plan of the technical contents, work breakdown into manageable tasks for each group member, and work schedule and budget justification.

Phase 2: Design: carrying out the design work as scheduled in Phase 1, documenting the design, preparing the design review, and updating the schedule and plan.

Phase 3: Prototype implementation, testing and troubleshooting. Ordering parts, components and other project-related materials.

Phase 4: Final design implementation and testing. Completion of the design and implementation work as scheduled in Phases 1 and 2, verification and testing if necessary, preparation of the final design review, technical and user manuals.

At the end of Phase 1 as well as Phase 2, students are required to submit progress report and make oral presentation. At the end of Phase 3 students are expected to present a working prototype of the system. Phase 4 is completed with an in-lab demo, a final and comprehensive project report, and a presentation in a poster/demo session.

References

Books

- P.G. Dominick, et al., Tools and Tactics of Design, Wiley, 1st ed., 2000.
- K. Smith, Teamwork and Project Management, McGraw-Hill Education, 4th ed., 2013.
- C. Selinger, Stu_ You Don't Learn in Engineering School, Wiley-IEEE Press, 2004.
[Available on IEEExplore]

Guides and Standards

- Guidelines to Professional Practice by the OIQ, 1999. ISBN: 2980218618. The last English version (maybe outdated in some parts):
<http://numerique.banq.qc.ca/patrimoine/details/52327/2217316>

Up to date official online version (in French only):

<https://www.oiq.qc.ca/membres/guide-de-pratique-professionnelle/>

Direct link:

<https://gpp.oiq.qc.ca/#t=Accueil.htm>

- IEEE 315-1975 (Reaffirmed 1993) Standard (available at the library through IEEE Xplorer)
- IEC 60617 Standard

Project Selection

In general, there are three sources of projects:

- 1) Faculty member proposed project.
- 2) Student(s) proposed project.
- 3) Industry proposed project.

Let us take a closer look at these three types of projects.

Faculty member proposed project.

The list of projects is currently available at Projects List web-page:

<https://users.encs.concordia.ca/~eceweb/capstone/projects-list.php>

This list is updated until the project selection deadline, so make sure to check it regularly. Projects, posted on the ECE web site, will be assigned on first come first served basis. Available projects have status “Available”, projects, which are already taken, – “In progress”. Professors may examine the qualification of the students, who intend to do the projects, and then decide if he/she will supervise them.

Student(s) proposed project.

Students can have their own proposal on a project, providing that a faculty member in the Department will supervise the project, solely or jointly. The project must be pre-approved by the Technical Coordinator, who will check the project scope, its feasibility, availability of resources to support the project, safety aspects, workload distribution among team members, etc. The workload should be sufficient for a team of six students.

If you wish to propose your own project, there are a few steps to follow:

- i) Send to the Technical Coordinator (dmitry@ece.concordia.ca) the following documents:
 - Project description, including the deliverables. You can submit more than one project idea, and we will discuss, which one would be the best for you.
 - Your assessment of resources required to complete the project.
 - General tasks distribution among team members.
- ii) Request for an appointment to review the proposal.
- iii) After successful revision, write a formal project proposal, and send it to the Academic and Technical Coordinators for final approval.
- iv) After your project proposal is approved, Academic coordinator will recommend it to a supervisor. Note that students are not required to approach faculty members with their project proposals.
- v) After the supervisor accepts the project, it will be posted on Projects List web-page; its status will be “In progress”. Note that this is informal assignment to prevent other teams from taking your project. See the instruction below on how to make your assignment final.

Industry proposed project.

If, for example, after your internship, your company wants you to continue to work on your project or give you a more complex one, or you have a contact in a company that has a relevant project to do, we can discuss it. Industry-inspired projects are the best option for you: you get wisdom of both academic and industry supervisors, and you get exposed to the cutting-edge technologies and their applications.

In general, the process is the same as for students proposed projects. However, there are a

couple of things that are different. First, the industry supervisor or representative has to attend the proposal revision meeting. Second, those projects might be subject to Non-Disclosure Agreements (NDA) between the students and the company. NDA terms and conditions will be discussed at the meeting with the company's representative.

In the last step, all team members should fill out and sign Project/Group Selection Form, get the signature of the supervisor, and submit it to the Academic Coordinator.

Projects will not be formally assigned to a group until the Academic Coordinator signs the Project/Group Selection Forms.

Grading Scheme

Please see the course outline.

Grading check points

- Project content, especially technical, and its scope.
- Project execution and management.
- Time management.
- Project results.
- Team: creation & management, task distribution, communications, etc.
- Reports and documentation: content, formatting, use of technical terms and standards.
- Presentations.
- Learning experience.

Graduate Attributes

Please see a separate document.

Zero Tolerance for any Kind of Academic Misconduct

Students should be familiar with University Code of Conduct (Academic) found at <https://www.concordia.ca/about/policies/spotlight/academic-code-of-conduct.html>

Each student needs to sign a Form of Expectations of Originality for all the written work of the course to certify that each submission is the original work of the group members and meets the Faculty's Expectations of Originality. The form is available at:

<https://www.concordia.ca/ginacody/students/academic-services/expectation-of-originality.html>

Please make sure that you sign the form **after** reading **the whole** page. The form must be signed **by hand in ink**. No electronic or any other kinds of signature are accepted.

Contacts

1. Academic inquiries, project assignment:

Academic Coordinator: Dr. Bahareh Goodarzi,
Email: bahareh.goodarzi@concordia.ca

2. Resources and technical support

Technical Coordinator: Mr. Dmitry Rozhdestvenskiy
Office: H851-2
Email: dmitry@ece.concordia.ca
Phone: (514) 848-2424 ext. 3106

3. General questions, presentation scheduling, special events

Program Assistant: Ms. Gulzat Temirova
Office: EV5.144
Email: undergrad-program-assistant@ece.concordia.ca
Phone: (514) 848-2424 ext. 3102

Project phases

General information on work submission

Report submission deadlines and presentation schedules: refer to the “Capstone Important Dates” web-page at the Department’s web-site:

<http://users.encs.concordia.ca/~eceweb/capstone/index.php>

Report

Guidelines

- For organization, layout and documentation please follow the guidelines from the Chapter 2-7 of the Concordia Form and Style Guide.
- Schematics, block diagrams, flow charts and any other graphical materials **must comply** with IEEE 315 or IEC 60617 standard.
- For groups doing software projects, you are permitted to use the ANSI/IEEE Standard for Software Requirements Specification, in place of the Concordia Form and Style Guide.

Submission

- All reports will be submitted electronically. The submission links will be available on Moodle.
- The teams should e-mail copies of their reports for each phase to the supervisors separately.
- There will be a 10% of full grade penalty per day for late submission of reports.

Presentation

- All members of each group should attend their group's project presentations. We also expect that students make themselves available for the entire hour block.
- Each presentation should be done by more than one student, preferably by all team members.
- Each group member is expected to talk at least 5 minutes cumulatively in the presentations (Phase 1, Phase 2, Phase 3 and Phase 4).
- The Phase 1 and Phase 2 presentations will take place via Zoom. All team members will be able to share their screens, but it is suggested that the team nominates a person who will be showing her/his screen and scrolling through the slides.
- Students are required to have their cameras on during the presentation and Q&A period.
- Each group has to book a time slot that is suitable for all the group members **and their supervisor** for their presentation.
- This time slot has to be given to Gulzat at least week before the presentation week.

Phase 1

- Create the team.
- Define the project requirements.
- Prepare the design specifications.
- Define technical contents.
- Break the project down into manageable tasks.
- Allocate tasks to individual group members.
- Prepare the schedule.
- Prepare the budget.

Deliverables

Group Report (20 pages)

- The Phase 1 report is intended to be a detailed project proposal.
- The length of the Phase 1 report is expected to be in the range of 20 pages¹.
- The report should contain:
 - o Abstract
 - o Objectives/Requirements/Specifications. Use Guidelines to Professional Practice (OIQ): {3.1.1.-3.1.3., 3.2.2., 3.2.7.} (*From now on, the figures in { } refer to chapters, clauses and paragraphs from this document*)
 - o Design specifications (numerical)
 - o Measurement of success (numerical)
 - o Test plan
 - o Review of existing solutions
 - o Alternatives
 - o Project planning and schedule {4}
 - o Team formation {4.1.}
 - o Tasks breakdown, description and allocation for each team member {4.1, 4.4}
 - o Communication plan
 - o Contingency plan {3.2.5}
 - o Tools required
 - o Budget estimation {3.2.9}

A sample title page is included here in Appendix 7. Please use this title page for all project submissions.

Presentation (based on Phase 1 report, 15 minutes per group)

- This presentation should proof that you:
 - o understand what the project is about;
 - o clearly see the goals and the deliverables;
 - o can justify that the project will be successfully completed;
 - o can properly manage the time and the scope of the project;
 - o have necessary qualifications/competences to complete the project.

¹ If your supervisor has any specific requirements, these should be addressed in this report.

- Each group is allotted 30 minutes: 15 minutes for presentation, 10 minutes for question and answer session, and 5 minutes for set-up and log-out. In general, you should provide an overview of your project and present your plan. You can also present any special requirements that your project may have.

Presentations FAQ

FAQ/FGC (frequently given comments) on your presentations and projects².

"If you can't describe what you're doing as a process, you don't know what you're doing" – Dr. W. Edwards Deming

1. What should a project report contain besides the technical aspects of the project?

1.1. As per the Project Management Body of Knowledge (PMBOK) the five basic process groups are:

- 1.1.1. Initiating.
- 1.1.2. Planning.
- 1.1.3. Executing.
- 1.1.4. Monitoring and Controlling.
- 1.1.5. Closing.

Therefore, "Phase 1" report should contain information on initiating and planning of your project. While planning your project, you should cover the following knowledge areas:

- 1.2. Project Scope Management.
- 1.3. Project Time Management.
- 1.4. Project Cost Management.
- 1.5. Project Human Resource Management (Team formation and management, tasks assignment).
- 1.6. Project Communications Management (Communications plan, meeting schedule, etc.)
- 1.7. Project Risk Management (Define negative and positive risks and the way you will be managing them).

Optional:

- 1.8. Project Quality Management.
- 1.9. Project Procurement Management.
- 1.10. Project Integration Management.

2. Specify project goals and clearly list deliverables at the beginning.

Example:

Project name: Fly-chasing machine.

Goal: Build a fly-chasing machine.

Deliverable (s): a working prototype of fly-chasing machine.

Design requirements

- Functional requirements:
 - o eliminate flies within certain area;
 - o log the parameters of the flying habits of the flies.
- Non-functional requirements:

² Note that not all comments are applicable to research projects

- o portable;
- o dual power: standard AC outlet and car battery;
- o suitable for outdoor use;
- o easy to setup and operate;
- o safety, etc...

Design specifications³:

A clean template of the specifications table is available on Moodle. Get it and use it.

Step 1: What the customer wants and how he sees the problem.

Unless otherwise noted (specified): ambient temperature is +20C, 60% RH, no direct sunlight. There are no obstacles within 15 m from the device. No wind. No precipitation. The device is powered from a standard power outlet (V=120V+-5%, F=60+-1 Hz). The size of the test flies is L/W/H = 5/5/5 mm; colour is black, matt.

#	Description/parameter	Test conditions	Value			Unit	Comments
			Min.	Typ.	Max.		
1.	Detection range	//if nothing is written here – the default test conditions apply	0			m	Farther it detects is better.
2.	Tracking speed		20			km/h	Higher speeds it can detect is better.
3.	Hit range		0			m	Farther it can hit the target is better.
4.	Number of flies tracked simultaneously	Normal ^{Note1}	5			pcs	More flies it can track is better.
5.	Performance	Normal ^{Note1}	3			flies/minute	More flies it can hit is better.
6.	Size (L/H/W)				200/ 200/ 100	mm	Smaller size is better.
7.	Power requirements ⁴ : ▪ AC source ▪ Voltage ▪ Frequency ▪ Current ▪ DC source ▪ Voltage ▪ Current	V=12V	10.5	120 60	0.5 24 2	V Hz A V	It is a bit too specific, but we need to figure out what kind of the power source for our system is acceptable by the client.
8.	Operational temperature		0		40	C	
9.	Storage temperature		-20		80	C	
10.	Relative humidity	Non-condensing			95	%	

Note 1: Normal test conditions mean that the test is conducted at ambient temperature +20C, 60% RH, no direct sunlight. There are no obstacles within 15 m from the device. No wind. No precipitation. The device is powered from a standard power outlet (V=120V+-5%, F=60+-1 Hz). The size of the test flies is L/W/H = 5/5/5 mm; colour is matt black (it is here just to show you how it could be done. You do not have to repeat the default test conditions once you listed them at the top of the table).

Usually, the general test conditions are put at the top of the table – they are applicable to

³ Software projects are NOT exempt from specifications

⁴ Might not be applicable to all projects. It might be a part of the design decisions.

all parameters, no need to reference each of them, i.e. to mention “Normal^{Note1}” in the “Test conditions”. Those are the default test conditions.

If you have many different test scenarios, you might want to limit the general test conditions to bare minimum, e.g. the temperature, and the rest of test conditions specify in the corresponding cells.

Notes can be used when the same test condition is applicable to many parameters.

Step 2: How we would rewrite/link what the customer wants to the real world (feasibility study) – our final design specifications.

Unless otherwise noted (specified): ambient temperature is +20C, 60% RH, no direct sunlight. There are no obstacles within 15 m from the device. No wind. No precipitation. The device is powered from a standard power outlet (V=120V+-5%, F=60+-1 Hz). The size of the test flies is L/W/H = 5/5/5 mm; colour is black, matt.

#	Description/parameter	Test conditions	Value			Unit	Comments
			Min.	Typ.	Max.		
1.	Detection range		.05		5	m	The detection range cannot be infinite: - there are obstacles; - there is no need to eliminate the flies if they don't approach your
2.	Tracking speed	Normal ^{Note1}	0		5 —	m/s	No system can track a subject flying with the speed of light. Let's see what we are dealing with: flies do not fly faster than that (<i>feasibility study-identify the speed</i>). “0” was removed from “Min.” because if the system can track 5 m/s, it should be able to track slow-moving objects (not always true, e.g. Doppler radar).
3.	Hit range	Normal ^{Note1}	0.1		3	m	No system can hit a target outside of its detection range. Plus, we don't have to chase “casual entrances” into the protection zone.
4.	Number of flies tracked simultaneously	Normal ^{Note1}	5		6	pcs	No system can track infinite number of targets. However, in that particular case, we might keep just “5” as our minimum requirement.
5.	Performance	Normal ^{Note1}	3		5	flies/minute	More flies it can hit is better.

6.	Size (L/H/W)				200/ 200/ 100	mm	Smaller size is better.
7.	Power requirements ⁵ : - AC source - Voltage - Frequency - Current - DC source - Voltage - Current			120 60	0.5	V Hz A	It is a bit too specific, but we need to figure out what kind of the power source of our system is acceptable by the client.
		V=12V ^{Note2}	10.5		24 2	V A	
8.	Operational temperature		0		30	C	
9.	Storage temperature		-20		80	C	
10.	Relative humidity	Non-condensing			95	%	

Note 1: Normal test conditions mean that the test is conducted at ambient temperature +20C, 60% RH, no direct sunlight. There are no obstacles within 15 m from the device. No wind. No precipitation. The device is powered from a standard power outlet (V=120V+-5%, F=60+-1 Hz). The size of the test flies is L/W/H = 5/5/5 mm; colour is matt black (it is here just to show you how it could be done. You do not have to repeat the default test conditions once you listed them at the top of the table).

Note 2: The consumption power should not exceed 48W for any type of voltage source.

Measurement of success: 90% of flies eliminated within 1.2m from the device, given that no more than 2 flies per minute enter in the 2 m space around the gadget and no fly leaves it.

IMPORTANT: For each number you write you must to have an idea how to prove that your design has met it. In other words, you must have an idea how you will write a **test case** for it.

3. **Analyze the market:** is there a similar product/solution? If so, explain why and what you are going to improve. If you are planning to redesign an existing product – just say so.

4. **Alternatives analysis**

4.1. Define your key parameters. This is applicable to each and every stage when you deal with alternatives (solutions, approaches, hardware, etc.).

4.2. Pick 2-4 alternatives.

4.3. Analyze alternatives using numbers instead of words, e.g.:

Parameter	Importance	Alternative 1	Alternative 2	Alternative 3
Key parameter 1	95	7	10	9
Key parameter 2	90	9	5	10
Key parameter 3	88	6	7	10
Parameter 1	60	10	2	1
Parameter 2	40	7	10	0
Cost	-50	9	10	8
Total score:		24.33	20.36	22.95

As you can see, “cost” is a negative parameter. You might have to add other “negative” parameters (e.g. effect on the environment).

If you mention numbers – be ready to explain how you got them (e.g. calculation,

⁵ Might not be applicable to all projects. It might be a part of the design decisions.

article, etc.).

5. **What is the difference between Phase 1 and Phase 2 alternatives analysis?**

Phase 1 - SYSTEM level ("big picture"), phase 2 – system COMPONENT level.

Example: your task is to design a machine that cuts trees. Thus, based on the size of the tree, cutting time, environment, etc. (i.e. project/design requirements) you have to make a choice of tools: chain saw, axe, knife, hand saw, etc... That's Phase 1. Let's say, you've picked an axe. In Phase 2, you have to decide on the blade and handle material and shape; how to sharpen the blade, etc...

6. **Cost estimation.** At the first stage, usually, you do not know what components you will be using, thus you cannot calculate cost based on the parts list. Instead, you can search the market and find the similar products. Your project cost estimation should be based on this information.

7. If you want to improve an existing product, do cost/benefits analysis.

8. **Deliverables** should be clearly stated for each phase.

9. Clearly state what you are **going to show** at the final presentation.

10. **Feasibility study** should justify that the project is doable within given time and budget.

11. **Do not omit the following areas:**

11.1. Team formation and management. Justify each team member selection.

11.2. Communication plan.

11.3. Time management.

11.4. Risk management. //consider team and project related risks; positive and negative.

12. **Use Guidelines to Professional Practice (OIQ):**

12.1. Objectives/Requirements/Specifications: {3.1.1.-3.1.3., 3.2.2., 3.2.7.}

12.2. Task Description/Allocation for each group member: {4.1., 4.4.}

12.3. Project planning: Chapter 4.

12.4. Team formation: {4.1.}

13. **Slides**

13.1. **Number the slides! Don't number the title page(s).**

13.2. Use tables, graphs and sketches instead of plain text.

13.3. Use legible color schemes and font size.

13.4. Put on the slides only necessary information.

Phase 2

- Design.
- Document it.
- Carry out the design work scheduled in Phase 1.
- Prepare the design review.
- Update schedule and plan (final deliverables evaluated against the revised plan).

Deliverables

Group Report (15-20 pages)

- Abstract.
- Review the design specifications {3.2.11, 3.3.14}.
- Present a design review (go over what has been done up to this point). Use necessary graphical materials (block diagrams, flow charts, graphs, etc.).
- Alternatives {3.1.2}:
 - list and analyze;
 - clearly describe trade-offs;
 - justify your choice (use numbers!).
- Developing the solution selected {3.1.4}.
- Design development {3.2.4}.
- Discuss and justify deviations from the initial task allocation/schedule.
- Update the schedule. Show how you addressed the deviations.
- Each student submits 1-2 page summary of his/her contributions.
- Appendix A: The first ELSEE assignment (see the capstone ELSEE Moodle page for information).
- Appendix B: The "Life-Long Learning" form is posted on Moodle under "Forms". Each student must complete a copy of the form. All forms must be included in an appendix in the capstone phase 2 report.

Presentation

- 40 minutes per group: 20 minutes for presentation of the work, 15 minutes for questions and answers, and 5 minutes for set-up and log-out.
- Those team members, who did not present the Phase 1 work, must make presentation for Phase 2.
- The presentation should be based on the report.
- Present your main design/implementation results in Phase 2 and the problems to be solved in Phase 3.
- Justify the modification of your schedule and project scope, deliverables, specifications, etc. given in Phase 1 if applicable.
- A working demo of a part of your project is encouraged, but not mandatory.

Phase 3

- Design implementation.
- Components and other project related materials procurement.
- Troubleshooting.
- Carry out verification and testing.

- Prepare the documents for final design.

Deliverables

In-lab demo

Each group presents its project in an ECE lab room to demonstrate the function and functionality of the prototype of so far implemented system. The group needs to book a time slot for the demo. **Make sure that, as usual, your supervisor is available.**

Phase 4

- Final design implementation.
- Complete the design and implementation work as scheduled in Phases 1 and 2.
- Carry out verification and testing.
- Prepare the final design review and final project report.
- Prepare technical and user manuals.

Deliverables

In-lab demo

Product conformance and final acceptance. Each group presents its project in an ECE lab room to demonstrate the function and functionality of the developed and implemented system. The supervisor, academic coordinator and technical coordinator will run comprehensive tests on the system to verify, if the system meets the requirements and specifications that were finalised in Phase 2. Internal signals, variables, and other parameters can be checked as well. Verification and testing will be done with hardware and software tools available in the lab. The group needs to book a time slot for the demo. **Make sure that, as usual, your supervisor is available.**

Group Report

This report is a complete report of the capstone project. It covers the entire development of the project, not only the work done during the Phase III. In this report, the description of the system developed should be presented, and the results obtained and the analysis/assessment of the results should be highlighted. Generally, the following sections should be included in the report.

- Abstract.
- Introduction (objectives and outlines of the project).
- Description of the project (design, development, problems encountered, solutions proposed, ...)
- Results obtained.
- Conclusion.
- References.

The number of pages of the report should not exceed 50. The description should be concise. The results should be presented with enough details (e.g. conditions for the tests) so that one

can understand easily how you got them and how to use your design/product. To make the report complete, the following points should be presented.

- Final **product** specifications.
- Description of the design and test results. All the necessary schematics and diagrams should be included, but no need to include the whole listing of your software, unless it is required for better understanding of your system's functionality. A CD/DVD with the project-related materials could be attached, as appendix, to the report.
- "Sales pitch" of the design.
- Implementation of the design. The description of what you have done and discuss the reasons for any deviations from the initial schedule submitted in Phase 1 and the revised one in Phase 2 should be presented.
- Conclusions.
- Feedback on the lessons learnt.
- Technical manual and the user manual of the system that you developed. They are to be attached as appendices.
- Appendix A: The second ELSEE assignment (see the capstone ELSEE Moodle page for information).

As an option, you may attach materials relevant to your design, as appendices, to your report. However, an attachment of excessive material is NOT encouraged (e.g. do not attach complete datasheets). All submitted materials should be bound together. No piece separated from the report will be accepted.

Presentation (final poster/demo session)

- All the members of each group are expected to be present at the poster/demo session, responding to questions from visitors, faculty, students, and examiners.
- The detailed scheduling of the sessions will be announced by mid-March.
- Each group will have a poster board of 4' by 8' (1.12m by 2.43m) (see "*Poster session plotting and preparation*")
- The contents of the poster should cover, in a brief and concise manner, the following aspects of your project:
 - The objective(s) of the project.
 - The development of the project.
 - The problem(s) encountered and the solution(s).
 - The results obtained.
 - The conclusion.

Individual Contribution Report

- This report is to be submitted by every student.
- You are expected to rate the contribution of each of your group members, and provide your point of view to justify your rating.
- This report is compulsory and it will be used, at the end of your project, by your project supervisor to assess the individual members' contribution to the work of the project.

Final Capstone Poster/Demo Presentation

The final presentation takes place at the end of March or beginning of April, usually between March 20 and April 10, in EV building atrium.

Usually, there are two sessions: morning and afternoon. The tentative schedule for these sessions is, if not otherwise specified, as follows:

First (morning) session:	Second (afternoon) session:
Setup: 9:00-9:30	Setup: 13:00-13:30
Presentations: 9:30-12:30	Presentations: 13:30-16:30
Dismantlement: 12:30-13:00	Dismantlement: 16:30-17:00

What will be provided and what is available (if not otherwise specified):

1. By default:

- Table (32'x100' – Area_1 or 30'x60' – Area_2).
- Power connection (AC 110V@15A).
- Poster board (4'x6', HxW). You can place up to two 36"x24" posters on it.

2. Upon request:

- Any piece of equipment from the project rooms (H857, H803 and other).
- PC with standard ENCS lab software.
- PC that the team used in the project room.

3. NO projectors and/or audio-video presentation equipment will be provided. If you wish to have it, please make necessary arrangements with IITS (Hall building, 4-th floor).

4. LAN. Only wireless network will be available. Make sure that you have a valid Alcor account. NO ENCS wired LAN will be available.

The following information should be provided by the team leaders:

- Team's availability for the both sessions. Indicate the number of students available for each session. (You have to do it, even your team has conflict with only one session.)
- If one or more team member(s) is(are) NOT available for a session, reasons **must** be given for each team member unavailable at the specified time(s). Note that only academic conflicts will be taken into consideration. In such case, please specify the course number.
- Equipment requirements. You can request ANY equipment from H-857, including the PC assigned to your team. (Please specify: model number and the ENCS tag number).
- Space requirements. If you need extra space (ex.: ground for robots), please specify desired size and shape, and justify the need.

5. Location requirements. If you wish to have a nice view from the window behind you, please specify the desired location and justify the need (ex.: bulky equipment located on the street, a camera pointed at the specific point, etc.). To get an idea of what is available see the detailed floor plan (Fig. 1).
6. If you will be using your own laptop, you can ask for an LCD monitor.
7. LAN. If you request a desktop PC, and you are planning to connect it to the network, please indicate it. Otherwise you will not have a wireless LAN card installed in your PC.

Any other requirements will be considered on per-case basis. Such requirements must be approved **before** the deadline.

Poster session plotting and preparation

The poster session is coming up and the projects will soon have the final presentation performed.

There will be an announcement for the scheduling of plotting during the last two weeks before final presentation. There will be approximately 150 groups trying to plot their posters at the last minute. It is recommended that you try to avoid the rush time by doing what you can before these two weeks. No project drawings will be allowed during these two weeks except by special permission. Poster size is recommended to be 24" X 36" and saved as a PDF file.

Also, remember that Capstone students have access to reserving room H821 for practicing and working on their presentation with a projector. You can request that the room be exclusive to your group during the reservation time. So far the room has not been heavily utilized by Capstone teams but I expect the request will be more forthcoming as the due dates near. Contact me to arrange the reservation time. The available times are posted outside the door.

Capstone Plotters Booking Procedures

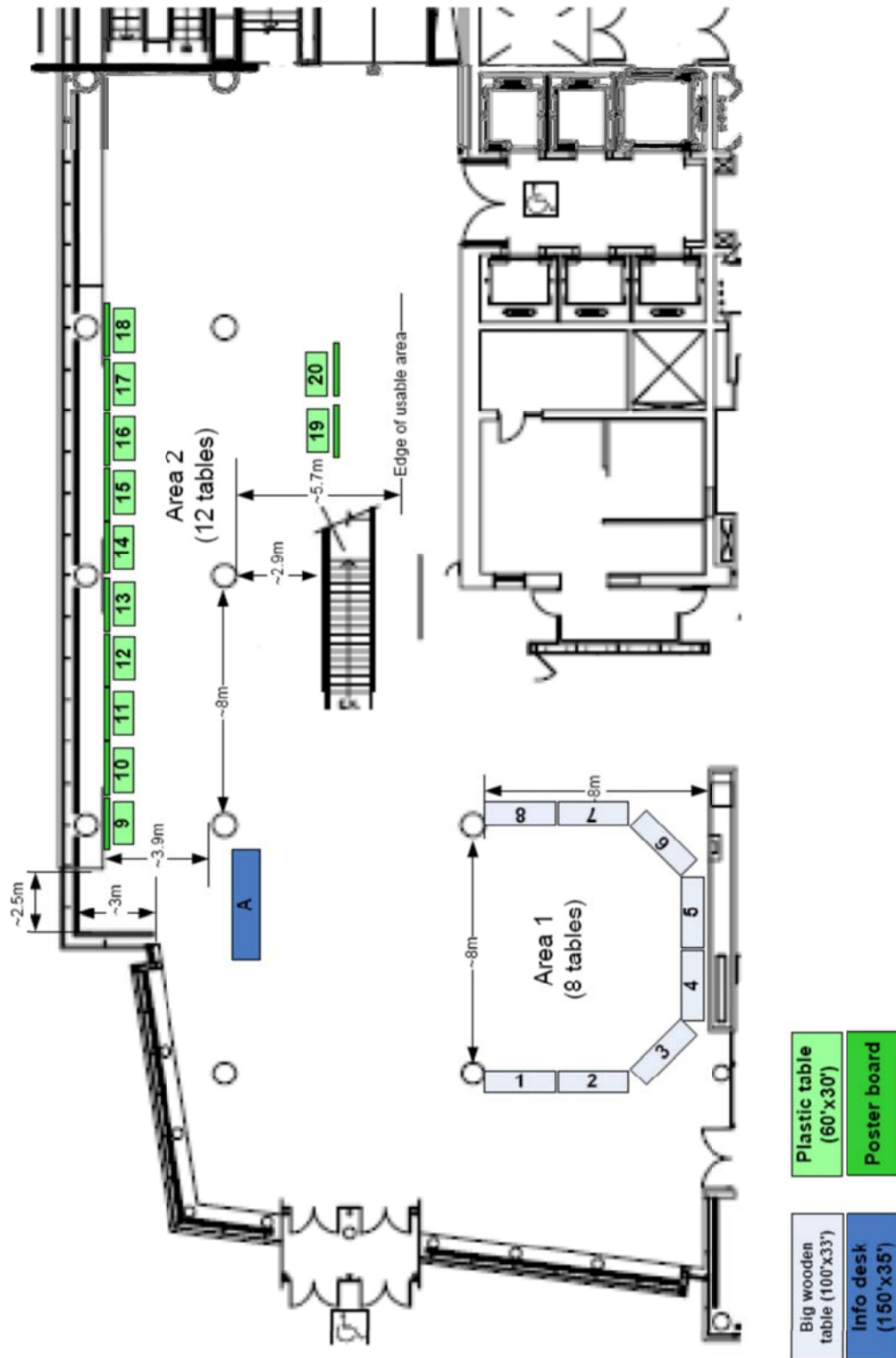
Since usually we have 60-90 teams and want to advantage all teams, we need to place limitations on plotting times and sizes.

The plotters are managed by the AITS. Please follow the procedures described here:

<http://www.concordia.ca/ginacody/aits/plotting-capstone.html>

Please note that the plotters get very heavily booked a day or two before the presentation date. So, make sure to make a reservation in advance.

Guy Street



St.Catherine

Figure 1: Tentative sample of final presentation floor plan.

Resources

Tech support structure

Technical Coordinator

Rozhdestvenskiy, Dmitry.

Specialists:

Li, Dan – real-time, control and embedded systems, web.

Obuchowicz, Tadeusz – digital systems, VHDL, VLSI.

Qin, Shiyu – analog systems, microelectronics. Safety officer.

Patel, Bipin – software, including Android.

Technicians:

Voronko, Alexey – RF, microwave, general inquiries,

Moreau, Jorris –telecommunication, PCB assembly, parts, tools, equipment.

Brikho, Chafic – power labs, materials ordering.

Laboratory Consulting Services

The Technical Coordinator will be available for consulting services. Students should bring their project description, block diagram, electronic schematic and any other documentation related to their project.

Laboratory Resources

List of labs

The Department has a number of labs associated with the 490 project.

Hardware labs:

- **H857**
Main Capstone Hardware Lab. Available to all teams. Lots of equipment and tools. Access to the PCs in H857 is available upon request only. ENCS password will NOT work.
Materials and tools listed as “available upon request” will only be installed in the lab as needed. To obtain this equipment, please contact the Technical Coordinator specifying the equipment required and a description of its use. Appendix 6 contains the request form.
- **H803**
The second Capstone Hardware Lab. Available to all teams. The lab is equipped with modern equipment and tools.
Access to the PCs in H803 is available upon request only. ENCS password will NOT work.

Other rooms/labs:

- **General Purpose Teaching Labs H913. H915. H917. H919. H921. H831**

Software available in Windows on the User-Managed Desktop:

http://aits.encs.concordia.ca/aits/public/desktop/USER_Managed_Desktop/Windows/Software/User_managed_software.html

For students who need word processing, you can use the word processing lab (H509) that the university offers to all students.

The list of computer labs is available from

<http://www.encs.concordia.ca/encsit/public/top/labs/index.html>

If a borrowed equipment or material is lost or damaged the University has the right to ask for re-imbursement.

Hardware labs access, PC and Locker Resource Request Procedure

In order to get access to hardware project rooms (e.g. H-857) you have to **pass the Health and Safety quiz on Moodle**, complete the form in Appendix 5 and send it by email to the Technical Coordinator. The form is available on Moodle as well.

Filling-in instructions

Group Members

All members should be listed and the email addresses should be the Concordia ones. This is important so that all members will receive any communications concerning your resources.

Computer Request

There are 4 shared and 7 private dual boot computers. The rule for the room is “one computer per group”, a group using the shared computers can access any one or more of the 4 computers depending on the activity in the room but the private computer groups can use only their assigned computer.

The computers are user managed, so the team has administrator privilege on them. All the teams using the shared computers must be extra careful not to cause problems as this could affect the other teams as well. The shared computers are meant for groups that have occasional need for a computer while in the room, for activities such as browsing or downloading code into a microcontroller. Any modifications to the supplied software or installation of new software should be approved in advance.

The private computers are assigned to one group only for teams that need to do extensive work using a computer and that need to install software and do constant testing of their project. Adding administrator users, changing administrator passwords or changing CMOS passwords should be indicated. The teams are responsible for doing their own backups of their files and are strongly advised to do so. Do not eliminate or damage the operating system, even if you do not use it, during the poster presentations your computer may be used by another project group on the alternate presentation session.

Project Description

Include details of any resources you expect to need.

Inform me if you wish to install another version of LINUX than what is supplied. This will require an addition hard disk to be installed. Also inform me of any hardware you plan to have installed.

Locker Request (there are 8 large ones, that you may have to share, and 8 small boxes with a combination lock for one group per locker.)

IMPORTANT: Access to any project room will be given only after the whole team successfully passes the Safety test on the Moodle. The passing grade is 100%. The test will include questions from the University's policy VPS-6 "POLICY ON WORKING OUTSIDE THE HOURS OF OPERATION OR IN ISOLATION".

Labs availability

- **9:00-23:00** – when the University is open (normal hours of operation): by default, sign up for a workbench is required to access certain labs.
 - **9:00 - 23:00** – every day: upon team's request sent directly to me. This type of access covers statutory and other holidays when the University is closed. Check out the Academic Calendar for the dates.
 - **24/7: NOT AVAILABLE.**
 - Deadline for the requests: see announcements on Moodle.
- The request template is available on Moodle. Fill in your names, student IDs only and the room number.

ENCS and University policies

Remember that the labs are subject to the following ENCS and University policies.

AITs Policies summary:

<http://www.concordia.ca/ginacody/ait/policies.html>

ENCS Account Policy <http://www.concordia.ca/it/services/encs-user-account.html>

POLICY ON COPYRIGHT COMPLIANCE:

<https://www.concordia.ca/content/dam/concordia/docs/VPERSG-2.pdf>

POLICY ON COMPUTING FACILITIES

<https://www.concordia.ca/content/dam/common/docs/policies/official-policies/VPS-30.pdf>

POLICY ON WORKING OUTSIDE THE HOURS OF OPERATION OR IN ISOLATION

<http://www.concordia.ca/vpirsg/documents/policies/VPS-6.pdf>

License Agreements

Students are obligated to respect all the license agreements and copyright agreements concerning all the materials they use for the capstone projects, regardless the ownership of the materials.

Design approval

- Final design can be approved⁶ either by the **Supervisor** or **Technical Coordinator**.
- Approval must be **explicit and documented**: it is not enough just to send your design and BOM for approval, or receive a verbal approval.
- The design approver should CC the approval e-mail to the other parties that include the Supervisor, Academic and Technical Coordinators. **Students are responsible** to ensure that all parties get informed on time, i.e. **before the approval deadline**.
- The approval email should have the approved BOM and all other supporting documents (e.g. schematics, block diagrams, flow charts, etc.) attached. **The approval is not valid without the attachments.**
- If the design is **not approved** by the deadline, it will be reflected in the phase 2 grade.
- If the design (BOM) is not approved by **January 22, 2027, 17:00**, the team might not receive the materials on time. The Department will not pay for rush shipping.
- Teams that **met the design approval deadline** will be given **priority** when it comes to technical support.

Components Request Procedure

1. Finish your design. Draw the complete system block diagrams and schematic(s).
2. Present them to the Technical Coordinator, justify your solution and get his approval of your schematic.
3. Fill in the BOM and submit it to the Technical Coordinator.
4. Submit the signed by the Technical Coordinator BOM to the technician (Chafic Brikho, H-961-30).

Components Available from ECE

The Department maintains a supply of commonly used electronic components such as op-amps, LEDs, transistors, and standard value resistors and capacitors. There is an inventory of materials that were used in the past Capstone projects.

Other components are available at H-941 upon approval of your request.

If the components are available at the Department, the technician will prepare them for you.

If the components are not available, see the Components Purchase Procedures below.

FAQ (design/schematic presentation):

1) Q: What kind of documentation should we bring to the meeting?

A: ANY materials that will help you to explain to me how the system works and justify your choice of components. Usually, you'll need:

- full schematic of the system with properly labeled components (please bring hard-copies)
- parts list
- datasheet(s) (necessary pages, or you can show them on your laptops)

2) Q: Is formal presentation required?

A: No. It's a Q&A session.

⁶ "Approved" means that all parties (the Supervisor, and Academic and Technical Coordinators) have received the approval e-mail with the BOM and all supporting documentation attached. Verbal approvals or e-mails without the BOM and appropriate documentation attached will not be accepted.

3) Q: What are the requirements to the drawings (diagrams, schematics)?

A: They must be legible and comply with the standards. Pencil sketches are OK.

4) Q: Is it necessary that the whole team is present at the meeting?

A: No. Ultimately, one team member with the consent of the team leader can do the job. However, I can better assess each team members' contribution if they participate in the defense.

5) Q: What are the typical questions you ask?

A: All type of questions that will unveil the level of your understanding of what you have done. Examples:

- What alternatives did you consider and analyse while you were choosing this component, module, etc.?
- What is the function of this (I point at a specific component) part in your system?
- How did you get this specific value of the parameter of the part?
- What will happen if I increase or decrease the value by 10 times?
- If you connect an oscilloscope here (I point at a specific point), what will you see (the shape of the signal and the boundaries for a "good" signal)?
- When you measured signal here, you got xx value. Is this value good or not? Justify.
- What might be a cause of the wrong value/shape?
- Briefly describe how you are going to debug and troubleshoot the system (hardware and software, test cases).

Project-related materials acquisition procedure

All materials can be requested only after the **complete** system has been designed and the design has been approved by the Technical Coordinator or the Project Supervisor. We have lots of materials available at the Department, so you can get them pretty fast. If materials are NOT available at ECE, student should procure them following the procedures described below.

Students are responsible for determining the supplier, quantity, price, availability, parts and catalogue number for the components that they want. **{3.3.1, 3.3.6}** If the information is incomplete, or the Supervisor's or Coordinator's signature is missing on the BOM, the order will not be processed.

Students can visit the sites below to determine the availability of the components that they need.

The deadline for the initial BOM submissions: see the course timetable. Usually, it is Friday before the Phase 2 presentations week by 17:00.

The materials will be ordered in the following weeks.

General materials purchase (Cost < \$1000)

These are preferred suppliers for electronics components purchase:

Digikey: <http://www.digikey.ca>

Newark: <http://canada.newark.com>

Mouser: <https://www.mouser.ca>

Electrosonic: <http://www.e-sonic.com>

Ali-express: <https://www.aliexpress.com/> Make sure that the seller offers "14-day delivery on orders over US\$8". Other shipping methods might cause excessive delays.

Procedure

1. Search the preferred suppliers for the component and obtain the price, availability, parts and catalogue number.
2. Once all the information is obtained, fill out the BOM (see the lecture: "How to Bill of Materials").
3. Have the BOM signed by the supervisor (optional).
The supervisor confirms that the materials are required for your project and will be used without implication.
4. Have the BOM signed by the Technical Coordinator. This will imply that the purchase can be made within the current budget.
5. Once the BOM is approved by the Technical Coordinator, the students will receive further instructions on how the order(s) will be placed.
6. Once the material is received by technical staff, the students will be notified to come to pick it up.

The students SHOULD NOT place orders or purchase any course-related materials on their own.

High cost materials purchase (Cost > \$1000)

The purchase of all components whose cost is more than \$1000 should be done following the procedure below.

The purchase **must** to go through the University, which will issue a purchase order.

Procedure

1. Search for suppliers of the material. You will be required to submit more than one price quotation, unless you are dealing with a sole source supplier.
2. Obtain the quotes. Attach the quotes to the BOM. Note that the quotations from at least **two vendors** are required.

Make sure that you have:

- Full address of the supplier
 - Contact person
 - Telephone and fax number
 - Email address
 - Web site info
3. The quote should be valid for at least 30 days.
 4. Have the BOM signed by your supervisor (optional).
The supervisor confirms that the equipment is required for your project and will be used without implication.
 5. Have the BOM signed by the Technical Coordinator.
This will imply that the purchase can be made within the current budget.
 6. The Technical Coordinator will undertake the appropriate steps to place a purchase order.
 7. Once the material is received by technical staff, the students will be notified to come to pick it up.

Students are asked to be patient for these items, as the entire process for purchase orders might take up to 4-5 weeks and even longer.

Purchased components reimbursement

Currently, there is no procedure to reimburse any materials purchased by the students out-of-pocket.

If a student purchases any materials on her/his own, even after the design was approved, **he/she is solely responsible** for the cost of aforementioned materials – **the purchase will NOT be reimbursed**.

Appendix 1: Sample TOC for technical manual

1. Intro
2. Objective
3. Specifications
4. System block diagram + description of operation
5. Hardware
 - a. Components/blocks/etc. (i.e.: chassis, motors, etc.)
 - b. Schematic (DO NOT FORGET to ANNOTATE it!!!)
 - c. Components functions and parameters description
 - d. Debugging/troubleshooting (schematics, low level)
6. List of components/Bill of materials (see “How-to...”)
7. Calibration/verification/testing (refer to the specifications)
8. Software
 - a. Flow chart
 - b. Description

Take a look at the posted example (Moodle) – it’ll give you good ideas on how to word your descriptions and what to put in the document.

Appendix 2: How to estimate the final cost for the items shipped to Canada

When you order parts abroad, keep in mind that the final price will be quite different from what you see in the quote. You will be charged custom duties, taxes and a brokerage fee.

How to estimate the final price:

1) Convert the invoice item price to CAD. Do not include shipping charge!

2) Go here (2026 update):

<https://www.cbsa-asfc.gc.ca/trade-commerce/tariff-tarif/2026/menu-eng.html> and find the custom tariff for your item.

Most of items of our interest will fall into this category:

<https://www.cbsa-asfc.gc.ca/trade-commerce/tariff-tarif/2026/html/00/ch85-eng.html>

3) Add percentage shown in "MFN tariff" column to the value calculated in (1).

4) Add GST (5%)

5) Add QST (7.5%)

6) Add brokerage fee:

- Canada Post usually charge flat rate \$10+tax
- USPS – does it through Canada Post (\$10+tax)
- UPS⁷ – based on the item's value (<\$20 – free; \$20.01 to \$40.00 - \$7.00; \$40.01 to \$100.00 - \$19.45; \$100.01 to \$200.00 - \$29.00; \$200.01 to \$350.00 - \$44.25).
- FedEx – give them a call 1-800-GoFedEx (item's value-based charge).
- DHL and other carriers – in most cases, the charge is based on the item's value, and could be as high as 30% of it; it might be quite high. Make sure you check with the company PRIOR to purchase.

Note that taxes apply to the item price PLUS custom tariff (duty).

There are a few exceptions from this rule (ex.: low priced (<\$20) items; items of value <\$60 (TBC), declared as "gift").

Here is a nice final cost calculator:

<https://www.cbsa-asfc.gc.ca/travel-voyage/dte-acl/est-cal-eng.html>

⁷ As of August 22, 2011. For up-to-date rates please visit www.ups.com

Appendix 3: Typical design roadmap

1. Assess the Customer's Needs (Voice of Customer, VoC) {3.1.1}.
2. Do a feasibility study. //Some customer's needs might be not feasible due to various constraints and contradicting requirements.//
3. Write down the initial/target design specification based on VoC. //Alice should know where she wants to get to!//Note: it is NOT a guessing game. There is no room for ambiguity and such things as "by default" and "based on common sense".//
4. Present these specifications to the customer and get his approval. //Get confirmation that you've addressed all customer's concerns.//
5. Establish the "Big Picture": Define the main problem. //"BIG Black Box", inputs/outputs, etc.//
6. Break it down to smaller tasks that you think you could come up with solution for. Consider alternatives for the break-down (system architecture) //Block diagram//
7. Derive from (3) specifications for each smaller task, sub-system, module. //BEFORE you make your choice, you have to understand your needs. It applies to system/sub-system architecture as well.//
8. Find 2-3 alternative solutions for each "smaller task". //Alternatives must be REAL alternatives, i.e. they should be suitable to a certain degree for the job. Do not compare flies to elephants!// {3.1.2}
9. If you cannot find a solution for a task - break it down to smaller tasks and try to solve them. //
10. Look for existing solutions and think if you can use them as the "building blocks" of your project. //Don't reinvent the wheel!//
11. For each of these solutions compile a list of "pros" and "cons", and assign a weight/importance coefficient to each parameter. Note that the use of the same resources (ex.: a piece of equipment) for more than one task could be considered as a "pro". Don't forget to estimate the cost of each solution.
12. Pick up the best alternative - it is your final solution candidate - you are ready to defend it.
13. Defend it (meet with me and your project supervisor). Prove that your solution is the best fit for customer's needs.
14. If any improvements are suggested - consider them.
15. Repeat 7-14 for the next task.
16. Integration: When you are considering a solution for each task – keep in mind that you have to put the things together and see how well your solutions fit one another. //Here you'll see if (7) was done properly.//
17. If necessary, make adjustments.
18. Make a list of required parts and equipment.
19. Trade-off assessment: Deal with constraints in your parts/equipment request.
20. Implement your solution!

Appendix 4: Sample Title Page

Phase 1 Report: Project Proposal

Cable Modem Performance

Team # 1

A Report
Presented to
The Department of Electrical & Computer Engineering
Concordia University

In Partial Fulfillment
of the Requirements
of ELEC/COEN 490

by
John Stone ID: 1231231
George Clement ID: 3213212
Elaine Bourne ID: 4564566
Chris Kramer ID: 7897899

Project
Supervisor(s) Dr.
Newman

Concordia University
October 2026
ELEC 490 & COEN 490

Appendix 5: Capstone lab PC, Locker and Resource Request Form⁸

Team# _____ Project Name: _____

Supervisor(s): _____

Team Members

Name (#1 - team leader)	ID Number	e-mail commonly used, phone number

Project Description (include details of any resources you expect to need):

Test equipment request (oscilloscope, DMM, etc.):

Computer Request: ☐ Dedicated ☐ Shared ☐ None

WIN7 ☐ LINUX ☐ Other OS ☐ Other Software ☐ Hardware ☐ Extra Table Space ☐
specify:

Locker Request: ☐ Large ☐ Small ☐ None

Assignment to the room: _____ workstation: _____

has been authorized: _____ Date: _____

⁸ The form is available on Moodle

Appendix 6: How to Bill of Materials

“A bill of materials (BOM) is a list of the raw materials, sub-assemblies, intermediate assemblies, sub-components, components, parts and the quantities of each needed to manufacture an end product. No physical dimension is described in BOM”⁹.

The BOM's header should contain the following:

1. Company's name
2. Project name
3. Design module/subassembly name
4. Designed by (team's or individual's name)
5. Revision number
6. Date

There are a few **levels of BOMs**; two of them are widely used: **system level BOM** and **sub-system** (assembly, board, sub-assembly, module) BOM. A lowest-level sub-system BOM consists of only purchased components or sub-systems manufactured by external companies. A top level system BOM includes all materials (components, modules, sub-assemblies, etc.) that the system is built of. In the system level BOM a sub-system appears as a single BOM item; there is no need to expand it. If it is an item that your company manufactures, it is up to you to assign a part number (ex.: module ABC123). The “unit cost” should include all expenses components, labour, shipping, storage, etc.) associated with the item. (see the example of a BOM on the Moodle).

While BOMs are usually customised for the company's needs, there is a number of fields that should be present. Below are the fields normally found in the BOMs of the electrical and computer engineering projects.

1. **BOM Item number**: a consecutive number that represents the part's position in the list, i.e. 1, 2, 3 ... Later at the assembly stage the part could be referred by this number rather than the design reference number.
2. **Design Reference**: it's the component's label/reference from your schematic or wiring diagram. You should group the components of the same value and other parameters. Ex.: R45-R75; D4, D7 ... The components should be sorted in alphabetical order by the design reference number.
3. **Description**: A summary of the component's most important parameters. Ex.: “10 kOhms $\pm 1\%$ 0.125W, 1/8W Chip Resistor 0805 (2012 Metric) Automotive AEC-Q200 Thick Film”; “N-Channel 80 V 100A (Tc) 176W (Tc) Through Hole TO-220”. **Hint**: use Digikey's “Detailed Description” field.
4. **Value**: value of the component. Ex.: “47 kOhm”, “2N3456” respectively.
5. **Quantity**: Total number of parts listed in “Design Reference” column.
6. **Unit (of quantity)**: “pcs”, “ea”, “m”, “kg”, etc... (Common mistake: use of unit of value, e.g. uF or kOhm)
7. **Manufacturer**: refers to a component manufacturer. This and the following column **must be included** in the BOM.
8. **Manufacturer part number**: a component manufacturer part number, NOT a supplier part number (!!!).

⁹ Monk, Ellen; Wagner, Bret (2007). Concepts in Enterprise Resource Planning. Course Technology Cengage Learning. pp. 97–98. ISBN 1-4239-0179-7.

The following columns are necessary for calculating the cost of the BOM.

9. **Supplier:** The part's supplier. (Ex.: Digi-Key)
10. **Supplier P/N:** Part number according to the **supplier's** inventory list. (Ex.: "ATMEGA8L-8PU-ND", "P100BATB-ND"). Often, it is different from the Manufacturer part number.
11. **Unit cost:** Cost per unit. Note that unit cost might vary depending on the number of units of the BOM item, e.g. one resistor can cost 13 cents, 10 resistors – 17 cents (0.017 cent each), 100 resistors – 67 cents (0.0067 cent each)¹⁰.
12. **Extended (or Total) cost:** "Unit cost" multiplied by "Quantity". Very bottom cell of this column should contain the total cost of materials.

Other fields you might find in BOMs are:

- For PCB manufacturing, "Footprint" column is often found in BOMs, e.g. TH (through-hole), 0805, LQFP64, etc. (the items that should NOT be installed must be marked, usually "DNS")
- Revision number
- Stock/warehouse reference number (for YOUR company)
- Internal reference number
- Reference to the specific drawing or schematic
- Alternative Supplier
- Comments

Since there are many columns in the BOM, you might want to choose "landscape" page orientation and even stretch it over two or more pages.

Often, for calculating the PCB manufacturing cost, the board summary is generated. It includes:

- total number of BOM items
- number of BOM items to be installed
- total number of components
- total number of surface-mount components
- total number of fine-pitch components
- total number of through-hole components
- total number of surface-mount components to be installed
- total number of fine-pitch components to be installed
- total number of through-hole components to be installed

A BOM template is available on Moodle.

¹⁰ <http://www.digikey.ca/product-detail/en/MCR01MRTF1002/RHM10.0KCDCT-ND/2796414>